House

"Danish Housing Market Analysis: Trends, Pricing & Sales Performance Dashboard"

Project Objective

The objective of this project is to analyze the Danish housing market using historical sales data to uncover trends in pricing, sales volume, and regional performance. The dashboard aims to help stakeholders (buyers, sellers, investors, or real estate analysts) understand how factors such as region, property type, sales type, and time period influence housing prices and sales activity — enabling data-driven decisions in property investment, pricing strategy, and market forecasting.

Data Source - Google big Query

- Creating Free Google Cloud Account
- Loading Data into Google big Query &. Connecting BigQuery to Power BI
- Using SQL in BigQuery for Data Understanding & Transformations
- Understanding & Cleaning the data using Power Query Editor

Data Link -. https://drive.google.com/file/d/1PpuWPloKOtTau5yG7QkXP2fuoDpQdXK4/view?usp=sharing

DAX Queries

- **Age** = ABS(YEAR(Housing[date (MM/DD/YYYY)].[Date]) - Housing[year_build])
- **Offer Price** = (100*Housing[purchase_price])/(100-Housing[%_change_between_offer_and_purchase])
- **Average Price SQM** = AVERAGE(Housing[sqm_price])
- **Last 12 Month Sales** = CALCULATE(SUM(Housing[purchase_price]),DATESINPERIOD(Housing[date (MM/DD/YYYY)],MAX(Housing[date (MM/DD/YYYY)]),-12,MONTH))
- **Offer to SQM Ration** = DIVIDE(SUM(Housing[Offer Price]),SUM(Housing[sqm]))
- **TotalYTD Sales** = TOTALYTD(SUM(Housing[purchase_price]),Housing[date (MM/DD/YYYY)].[Date])
- **Sales by Region** = CALCULATE(SUM(Housing[purchase_price]),ALLEXCEPT(Housing,Housing[region]))

DAX Queries

- **Median Sales Price Change** =
```
Var CurrMedianPrice =
   MEDIANX(FILTER(Housing,
   YEAR(Housing[date (MM/DD/YYYY)].[Date])=
    YEAR(MAX(Housing[date (MM/DD/YYYY)].[Date]))),
     Housing[purchase_price])

 Var PreMedianPrice =
  MEDIANX(FILTER(Housing,YEAR(Housing[date (MM/DD/YYYY)].[Date])=YEAR(MAX(Housing[date (MM/DD/YYYY)]. [Date]))-1),Housing[purchase_price])

 RETURN
 IF(PreMedianPrice<>0, (CurrMedianPrice-PreMedianPrice)/PreMedianPrice,
 BLANK())
```

- **Units sold in latest Year & Quarter** = CALCULATE(DISTINCTCOUNT(Housing[house_id]), YEAR(Housing[date (MM/DD/YYYY)])= YEAR(MAX(Housing[date (MM/DD/YYYY)])) && QUARTER(Housing[date (MM/DD/YYYY)])=QUARTER(MAX(Housing[date (MM/DD/YYYY)])))

- **YOY_Sales_Growth** =
```
 Var CurrYearSales =
  CALCULATE(SUM(Housing[purchase_price]),
          YEAR(Housing[date (MM/DD/YYYY)])=YEAR(max(Housing[date (MM/DD/YYYY)])))

 Var Prevyearsales =
 CALCULATE(SUM(Housing[purchase_price]),
  YEAR(Housing[date (MM/DD/YYYY)])=year(max(Housing[date (MM/DD/YYYY)]))-1)
 Return
  IF(Prevyearsales<>0, (CurrYearSales-Prevyearsales)/Prevyearsales,BLANK())
```

Project Insights

Page 1. Market Overview

- Total 12-month sales reached 13bn, with 77 units sold in the latest year/quarter.
- Bornholm recorded the highest median sales price increase (~5%) among all regions, followed by Jutland and Fyn & Islands — while Zealand showed the smallest price growth, suggesting Bornholm may be an emerging or currently hot micro-market despite its small transaction volume.
- Offer Price vs Purchase Price shows a near-perfect positive correlation, indicating buyers are consistently paying very close to the asking price — a sign of a stable, low-negotiation market (or accurate initial pricing by sellers).
- YoY Sales Growth by Sales Type reveals a sharp divide:
- Auction sales grew +29% YoY — the only sales type showing positive growth.
- Regular sales and other sales both declined -21%.
- Family sales dropped sharply by -75%, the steepest decline — suggesting a significant pullback in intra-family property transfers, possibly due to tax/regulatory changes or shifting inheritance patterns.

Project Insights

Page 2. Sales Performance

- Zealand dominates total sales volume at 95bn, more than 5x the next closest region (Jutland at 81bn), confirming it as the primary revenue-driving region despite modest price growth.
- Bornholm contributes just 1bn in total sales — the smallest share — reinforcing that its price growth is likely driven by low volume/high-value outliers rather than broad market strength.
- Key Influencers analysis found that buyer age is a major driver of purchase price: buyers aged 1–16 (likely properties purchased on behalf of minors/trusts) and 69+ are associated with the highest average purchase prices (+443.2K and +437.5K respectively) — much higher than mid-age buyers (16–45, 45–69).
- Offer-to-SQM ratio is highest for regular sales (14.9K) and lowest for auction sales (11.1K) — auction properties are consistently priced lower per square meter, consistent with auctions typically involving distressed or below-market sales.
- Average Price per SQM by region: Zealand leads at 20.85K/SQM (35.6% share), nearly double Bornholm's 10.6K/SQM (18.1%) — confirming Zealand as the premium region on a per-unit basis, not just in volume.

Project Insights

Page 3. House Type Analysis

- Farms command the highest average price (2.7M) and the largest average size (196 SQM), yet have the lowest price per SQM (13.83K) — indicating farms are priced on total land/size rather than premium per-unit value.
- Apartments have the smallest average size (87 SQM) but the highest price per SQM (28.69K) by a wide margin — confirming apartments are the most expensive housing type on a per-square-meter basis, typical of urban density premiums.
- Offer price and purchase price are virtually identical across all property types (e.g., Farm: 2.7M vs 2.7M, Apartment: 2.4M vs 2.4M), reinforcing the earlier finding of minimal price negotiation across the market.
- Yield is inversely related to price per SQM: Farms show the highest yield (4.6%) despite the lowest SQM price, while Apartments show a lower yield (3.9%) despite commanding the highest SQM price — suggesting an inverse relationship between entry price and rental/investment yield.
- Inflation and interest rates are nearly uniform (~1.9–2.0% and ~1.4–2.1%) across all property types, meaning macroeconomic factors are not the differentiator — property type and location drive price variation far more than market-wide economic conditions.

Thank You!